

Exam 1

- This is a closed-note, individual exam. No external materials, devices, or collaboration are permitted.
- For short-answer responses, write clearly and concisely. Use terminology from class where appropriate.
- When asked to justify or explain, provide reasoning to receive full credit.
- If a question is unclear, please ask for clarification.
- Answer all questions and manage your time carefully.

1 Multiple Choice 2 points

Which of the following best describes the starting point of good sports analytics?

- ☐ Selecting the most advanced statistical model
- ☐ A well-defined, stakeholder-oriented question
- ☐ Gathering the largest dataset available
- ☐ Running exploratory visualizations first

2 Multiple Choice 2 points

An analyst notices teams with higher possession percentages tend to win more games and concludes that possession causes wins. What error are they making?

- ☐ Treating a prescriptive question as descriptive
- ☐ Confusing pattern recognition with causation
- ☐ Using the wrong visualization type
- ☐ Ignoring data reliability

3 Multiple Choice 2 points

A front office analyst is asked to recommend which free agents to sign based on projected impact. This is best described as:

- ☐ Descriptive
- ☐ Diagnostic
- ☐ Predictive
- ☐ Prescriptive

4 Multiple Choice 2 points

Exploratory analysis is primarily used to:

- ☐ Confirm causal claims
- ☐ Generate questions to test
- ☐ Deploy final decisions
- ☐ Eliminate uncertainty

5 Multiple Choice 2 points

Jersey numbers assigned to basketball players are best classified as

- ☐ Ratio
- ☐ Interval
- ☐ Nominal
- ☐ Ordinal

6 Multiple Choice 2 points

Which variable is best classified as numerical and on continuous

- ☐ Injury status
- ☐ Player position
- ☐ Jump height
- ☐ Game date

7 Multiple Choice 2 points

An analyst focuses entirely on the most easily accessible tracking data while overlooking the actual evidence needed to answer the coaching staff's core question. Which analytical trap does this scenario describe?

- ☐ Data dredging
- ☐ Streetlight effect
- ☐ Spurious correlation
- ☐ Simpson's paradox

8 Multiple Choice 2 points

Which of the following best describes how modern player tracking data differs from traditional box score statistics?

- ☐ Box scores are less accurate than tracking data because they are recorded manually
- ☐ Tracking data is higher in volume and frequency and introduces greater risk of error if validation is not conducted
- ☐ Box scores capture more dimensions of performance than tracking data
- ☐ Tracking data eliminates the need for focused research question because it measures everything automatically

9 Multiple Choice 2 points

Which data collection method is most context-rich but can be subjective?

- ☐ Wearable technology
- ☐ Lab-based metabolic measure
- ☐ Optical tracking systems
- ☐ Observational scouting

10 Essay 4 points

Name one ethical consideration mentioned in class (1 pt). Discuss how it is relevant to sports performance analytics and how it might influence how an analyst does their work (3 pts).

11 Multiple Choice 2 points

Which best describes *validity* in data collection?

- ☐ Consistency across repeated assessments
- ☐ Whether a measure captures what it is intended to capture
- ☐ Completeness of a dataset
- ☐ Absence of outliers

12 Multiple Choice 2 points

Which statistical operation is NOT valid for ordinal data?

- ☐ Counting frequencies
- ☐ Ranking from highest to lowest
- ☐ Computing a mean
- ☐ Identifying the most common category

13 Multiple Choice 2 points

Which example best illustrates a *reliability* issue?

- ☐ A GPS tracking system consistently overestimates sprint distance by 10% for all athletes.
- ☐ A survey question fails to measure athlete motivation accurately.
- ☐ A biomechanics analysis device is calibrated incorrectly and produces biased values.
- ☐ The same athlete reports different exertion level (RPE) to two coaches for the identical workout.

14 Essay 4 points

Why is it important to avoid causal language (such as "cause" or "lead to") when interpreting observational sports data, and what alternative phrasing should be used instead?

15 Multiple Choice 2 points

Which of the following is true about the historical context and outcomes of the Oakland Athletics' "Moneyball" story as outlined in your course materials?

- ☐ Bill James published his influential critiques in the 1970s, roughly 30 years before the events popularized as "Moneyball".
- ☐ The Athletics achieved ultimate championship victory, proving that data-driven payroll constraints guarantee a title.
- ☐ Because the Athletics' approach failed completely, teams abandoned data analytics for decades afterward.
- ☐ Widespread adoption of market inefficiencies permanently locks in a team's competitive advantage without needing further adaptation.

16 Fill in the Blank 3 points

A dataset has one row per athlete and each column is a summary of various measures of the athlete's performance. This format is

and is most suited for .

☐ long

☐ wide

☐ compact representation and calculations across columns

☐ creating visualizations and using statistical programs

17 Multiple Choice 2 points

A basketball team's overall season average suggests that Player A shoots a higher field goal percentage than Player B. However, when broken down by individual defensive schemes, Player B actually shoots better against every single scheme. What statistical phenomenon does this scenario illustrate?

- ☐ Streetlight effect
- ☐ Simpson's paradox
- ☐ Spurious correlation
- ☐ Data dredging

18 Multiple Choice 2 points

Anscombe's Quartet is most commonly used to illustrate:

- ☐ The importance of using the median over the mean
- ☐ Details that descriptive statistics can hide which visualization can reveal
- ☐ The difference between cross-sectional and longitudinal data
- ☐ What Steph's face would look like as a shot chart

19 Multiple Choice 2 points

A team's average (mean) points per game is 108, but their median points per game is 94. This most likely suggests:

- ☐ The data contains errors
- ☐ A few very high-scoring games are pulling the mean upward
- ☐ The standard deviation is low
- ☐ The data is normally distributed

20 Multiple Choice 2 points

Which is a key limitation of only using averages (mean, median or mode) to describe data?

- ☐ They cannot be used to compare groups
- ☐ They can hide outliers, distributional shape, and individual-level trends
- ☐ They require continuous numerical data
- ☐ They are only appropriate for longitudinal datasets

21 Multiple Choice 2 points

If an analyst wants to compare the consistency of performance of different position groups, they should compare:

- ☐ Mean
- ☐ Median
- ☐ Standard Deviation
- ☐ Range

22 Multiple Choice 2 points

Which of the following statements is true regarding the evolution of sports data?

- ☐ The digital measurement revolution marked the beginning of systematic measurement and manual analysis
- ☐ Modern sports analytics is exclusively focused on scouting and has largely phased out integration with sports medicine and coaching
- ☐ The digital measurement revolution significantly increased both the depth and breadth of available data through new measurement technology
- ☐ Outside analysts began challenging convention only after the modern era of sports analytics was fully established.

23 Multiple Choice 2 points

Why might an analyst use a proxy measure when investigating a sports science research question?

- ☐ Because direct measurement of the actual concept is unavailable, impractical, or too costly, requiring an observable stand-in.
- ☐ Because proxy data completely eliminates all sources of measurement error and bias.
- ☐ Because league and competition providers mandate that all internal metrics must use derived variables.
- ☐ Because proxy measures automatically guarantee high validity compared to human observation.

24 Multiple Choice 2 points

Which visualization is most appropriate for examining the distribution of vertical jump height across a squad?

- ☐ Bar chart
- ☐ Scatterplot
- ☐ Line chart
- ☐ Histogram

25 Multiple Choice 2 points

Which visualization is most appropriate for comparing the average fastball velocity across five different prospect pitchers?

- ☐ Scatterplot
- ☐ Bar chart
- ☐ Line chart
- ☐ Histogram

26 Multiple Choice 2 points

A team's GPS tracking system has a typical measurement error (reliability threshold) of $\pm 2\%$. An analyst measures an athlete's high-speed running distance and observes a 1% increase from the previous week. Based on the reliability of the device, what is the most appropriate conclusion?

- ☐ The athlete has successfully improved their high-speed running capacity by 1%.
- ☐ The observed change is smaller than the measurement error, so we cannot distinguish a true physiological change from random measurement noise
- ☐ The device completely lacks validity, so the data is systematically biased.
- ☐ The athlete is experiencing neuromuscular fatigue and needs to be rested immediately.

27 Multiple Choice 2 points

Which of the following describes a key drawback associated with using third-party vendors for sports data?

- ☐ They completely lack technological expertise or advanced tools
- ☐ They involve financial costs, unclear algorithms, and vendor dependence
- ☐ They restrict data collection strictly to small-sample academic contexts
- ☐ They rely entirely on subjective observer judgment and manual video coding

28 Multiple Choice 2 points

Which visual encoding method allows the most precise comparison?

- ☐ Volume with 3d shapes
- ☐ Color intensity
- ☐ Position along a common scale
- ☐ Angles

29 Multiple Choice 2 points

A coach glances at a chart for five seconds and immediately identifies the top three performers. This is most likely because the chart uses:

- ☐ A color palette with 10 different colors
- ☐ Pre-attentive attributes like position and enclosure
- ☐ A radar chart showing multiple dimensions
- ☐ A pie chart for part-to-whole comparison

30 Multiple Choice 2 points

An analyst wants to show the relationship between training load and injury risk across a dataset with data for 50 athletes. The most appropriate chart is:

- ☐ Stacked bar chart
- ☐ Line chart
- ☐ Scatterplot
- ☐ Box plot

31 Multiple Choice 2 points

Which chart type is best for showing a player's sprint speed across a 20-week training season?

- ☐ Bar chart
- ☐ Histogram
- ☐ Line chart
- ☐ Treemap

32 Multiple Choice 2 points

Which pre-attentive visual attribute is most appropriate for encoding unordered categorical data, such as distinguishing between different player positions on a chart?

- ☐ Color intensity
- ☐ Length
- ☐ Color hue
- ☐ all of the above

33 Multiple Choice 2 points

Which visualization is most appropriate for comparing the proportion of different shot types (e.g., 3-pointers, mid-range, and shots in the paint) taken across three different NBA teams?

- ☐ Histogram
- ☐ 100% stacked bar chart
- ☐ Scatterplot
- ☐ Bar chart

34 Essay 3 points

When examining a scatterplot to analyze the relationship between two numerical variables, what key features or patterns should you look for and describe? List and briefly explain two.

35 Multiple Choice 2 points

When evaluating the breakdown of win-loss outcomes across different venues (home vs. away) where the total number of games played differs, why is it preferable to look at proportions (relative frequencies) rather than raw counts?

- ☐ Raw counts completely eliminate the need for categorical variable encoding.
- ☐ Proportions allow us to make fair comparisons possible when group sizes or sample sizes are unequal.
- ☐ Proportions automatically convert categorical data into continuous numerical distributions.
- ☐ Raw counts can only be visualized using scatterplots.

36 Essay 8 points

A college football program uses self-reported sleep hours as a key variable in their recovery tracking. Self-reported measures can introduce issues of validity and reliability in analytics. Identify one concern related to each (validity and reliability) and explain how those concerns could affect the conclusions or decisions drawn from the analysis.

A sports scientist collects 40-yard dash times (in seconds) from 30 NFL draft prospects.

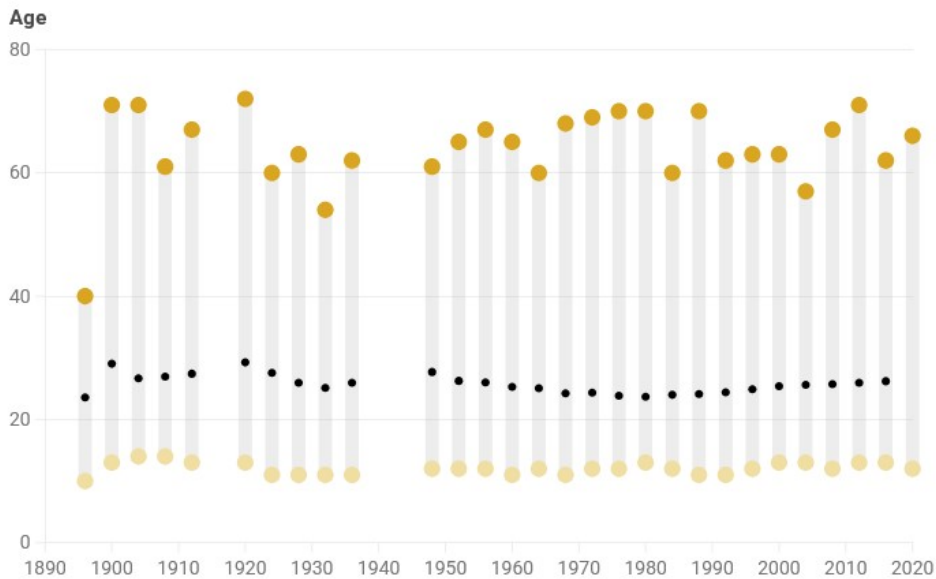
- What kind of data is this? Be more specific than "categorical" or "numerical". (1 pt)
- Write an analytical question that could be answered using **only** this dataset. (2 pts)
- Name one appropriate method for answering your question. (2 pts)
- Name and justify one appropriate visualization. (3 pts)

Examine the visualization provided.

- What does the x-axis represent? What does the y-axis represent? (2 pts)
- Identify two visual (pre-attentive) attributes used in the chart and describe what each encodes. (4 pts)
- What piece of information in this chart can be compared most precisely? Explain which visual attribute makes that level of precision possible. (2pts)

From 12 to 72: Ages of Olympians since 1896

● Youngest ● Oldest ● Average Age



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